

Day 65 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in TN Culture & Heritage (2): New GI Tags

READING MATERIAL

Covers Tamil Nadu's 5 newest Geographical Indication (GI) tag registrations and how the state ranks nationally — directly relevant to the 'Achievements of Tamil Nadu' and TN heritage clauses.

Q: What are the 5 new products that recently received GI tags in Tamil Nadu, and what is each?

A: (1) Woraiyur cotton sari (native to Tiruchi district); (2) Kavindapadi nattu sakkarai (a traditional jaggery powder); (3) Namakkal makkal pathirangal (soapstone cookware); (4) Thooyamalli rice (a special traditional rice variety); (5) Ambasamudram choppu saman (handcrafted wooden toys/items).

Q: With these 5 new additions, how many total GI-tagged products does Tamil Nadu now have, and how does this rank nationally?

A: 74 total GI-tagged products — ranking 2nd nationally, behind Uttar Pradesh (77) and ahead of Karnataka (around 45).

MOCK TEST (WITH ANSWERS)

1. Which of these is one of Tamil Nadu's 5 newest GI-tagged products?

A. Woraiyur cotton sari ✓

B. Kanchipuram silk sari

C. Madurai Sungudi sari

D. Coimbatore wet grinder

Explanation: Woraiyur cotton sari. [medium, web-research]

2. "Namakkal makkal pathirangal", one of TN's new GI-tagged products, refers to:

A. A jaggery powder

B. Soapstone cookware ✓

C. A traditional rice variety

D. Handcrafted wooden toys

Explanation: Soapstone cookware. [hard, web-research]

3. With its 5 newest additions, how many total GI-tagged products does Tamil Nadu now have?

A. 45

B. 65

C. 74 ✓

D. 90

Explanation: 74. [medium, web-research]

4. Which state currently has the highest number of GI-tagged products in India, ahead of Tamil Nadu?

A. Karnataka

B. Kerala

C. Uttar Pradesh ✓

D. West Bengal

Explanation: Uttar Pradesh (77). [medium, web-research]

Part B – Aptitude & Mental Ability: Simplification

READING MATERIAL

Simplification is 1 of 10 named Aptitude sub-skills (15 questions total for the whole Aptitude unit). TNPSC's simplification questions are rarely 'plug into one formula' — they combine BODMAS, fractions/decimals, ratios, and algebraic shortcuts. Each item below is a worked method, not just an answer — the goal is to recognize which technique a given expression calls for.

Q: Method — BODMAS with a nested bracket and an unknown: Solve for x : $57 - [42 - \{37 - (20 - x)\}] = 176 \div (8 \div 2)$

A: Work from the innermost bracket outward, and simplify the RHS first: $176 \div (8 \div 2) = 176 \div 4 = 44$. Now the LHS: $(20 - x)$ stays as is for now $\rightarrow \{37 - (20 - x)\} = 17 + x \rightarrow [42 - (17 + x)] = 25 - x \rightarrow 57 - (25 - x) = 32 + x$. Set equal to 44: $32 + x = 44 \rightarrow x = 12$. (Real TNPSC-style problem; this exact expression is a recurring coaching-site reference question.)

Q: Method — sum of two squares without brute-force squaring: Find $217 \times 217 + 183 \times 183$.

A: Use $a^2 + b^2 = (a+b)^2 - 2ab$ instead of squaring two 3-digit numbers by hand. Here $a+b = 400$ (round number — that's the clue this trick applies) and $ab = 217 \times 183 = 39,711$. So $a^2+b^2 = 400^2 - 2(39,711) = 160,000 - 79,422 = 80,578$. Recognizing 'the two numbers add to a round number' is the trigger for this shortcut.

Q: Method — difference of squares shortcut: Simplify $(197 \times 197 - 103 \times 103) \div (197 - 103)$ without long multiplication.

A: $a^2 - b^2 = (a+b)(a-b)$, so $(a^2-b^2)/(a-b) = (a+b)$ directly — no multiplication needed at all. Here $a=197$, $b=103$, so the answer is simply $197+103 = 300$. This 'difference of squares divided by the difference' shape is a favorite TNPSC trick question precisely because it looks like it needs heavy calculation but collapses to one addition.

Q: Method — fraction-of-decimal addition: Simplify $7/5$ of $58 + 3/8$ of 139.2 .

A: $7/5$ of $58 = 7 \times 58/5 = 406/5 = 81.2$. $3/8$ of $139.2 = 3 \times 139.2/8 = 417.6/8 = 52.2$. Sum: $81.2 + 52.2 = 133.4$.

Q: Method — ratio substitution to simplify a compound ratio expression: If $x : y = 3 : 4$, find $(4x + 5y) : (5x - 2y)$.

A: Don't solve for x and y separately — substitute $x=3k$, $y=4k$ (k is just a scaling factor that will cancel). $(4x+5y) = 12k+20k = 32k$. $(5x-2y) = 15k-8k = 7k$. Ratio = $32k : 7k = 32 : 7$. The k always cancels in a ratio question — that's the whole trick.

Q: Method — same ratio-substitution trick, different ratio: If $x : y = 3 : 5$, find $(5x + 3y) : (15x - 2y)$.

A: Substitute $x=3k$, $y=5k$. $(5x+3y) = 15k+15k = 30k$. $(15x-2y) = 45k-10k = 35k$. Ratio = $30k:35k = 6:7$.

Q: Method — simultaneous equations feeding into a simplification target: If $3a + 4b = 22$ and $8a - 5b = -4$, find $(a+b)^2$.

A: Solve the pair normally: from $3a+4b=22$ and $8a-5b=-4$, elimination gives $a=2$, $b=4$ (check: $3(2)+4(4)=6+16=22$ ✓; $8(2)-5(4)=16-20=-4$ ✓). The question doesn't ask for a and b individually — it asks for $(a+b)^2 = (2+4)^2 = 36$. TNPSC often hides the real target $((a+b)^2, \text{ not } a \text{ or } b)$ at the end of a simultaneous-equation question — always re-read what's actually being asked before solving for individual variables.

Q: Method — mixed numbers with a nested fraction: Simplify $2\frac{1}{2} + 3\frac{1}{3} \div \frac{2}{3}$.

A: BODMAS still applies to fractions — division before addition. $3\frac{1}{3} \div \frac{2}{3} = (10/3) \div (2/3) = (10/3) \times (3/2) = 5$. Then $2\frac{1}{2} + 5 = 7\frac{1}{2}$. The common mistake is adding $2\frac{1}{2}+3\frac{1}{3}$ first — division always binds tighter than addition, fractions or not.

MOCK TEST (WITH ANSWERS)

1. Solve for x : $70 - [45 - \{33 - (15 - x)\}] = 171 \div (9 \div 3)$

- A. 10
- B. 12
- C. 14 ✓**
- D. 16

Explanation: RHS: $171 \div (9 \div 3) = 171 \div 3 = 57$. LHS inner-out: $\{33 - (15 - x)\} = 33 - 15 + x = 18 + x \rightarrow [45 - (18 + x)] = 27 - x \rightarrow 70 - (27 - x) = 43 + x$. Set $43 + x = 57 \rightarrow x = 14$. [medium, authored]

2. Find $312 \times 312 + 288 \times 288$ using the $(a+b)^2 - 2ab$ shortcut.

- A. 179,808
- B. 180,288 ✓**
- C. 190,000
- D. 182,688

Explanation: $a+b=600$, $ab=312 \times 288=89,856$. $a^2+b^2=600^2-2(89,856)=360,000-179,712=180,288$. [medium, authored]

3. Simplify $(425 \times 425 - 375 \times 375) \div (425 - 375)$.

- A. 400
- B. 750
- C. 800 ✓**
- D. 425

Explanation: Using $a^2 - b^2 = (a+b)(a-b)$, dividing by $(a-b)$ leaves just $(a+b) = 425+375 = 800$. [easy, authored]

4. Simplify $\frac{5}{6}$ of 72 + $\frac{2}{7}$ of 98.

- A. 88 ✓**
- B. 84
- C. 90
- D. 86

Explanation: $\frac{5}{6}$ of 72 = 60. $\frac{2}{7}$ of 98 = 28. Sum = 88. [easy, authored]

5. If $p : q = 5 : 6$, find $(3p + 2q) : (4p - q)$.

- A. 27 : 14 ✓**
- B. 9 : 7
- C. 27 : 7
- D. 14 : 9

Explanation: $p=5k, q=6k$. $(3p+2q)=15k+12k=27k$. $(4p-q)=20k-6k=14k$. Ratio = 27:14. [medium, authored]

6. If $m : n = 4 : 7$, find $(2m + 3n) : (5m - n)$.

- A. 29 : 13 ✓**
- B. 13 : 29
- C. 29 : 20
- D. 20 : 29

Explanation: $m=4k, n=7k$. $(2m+3n)=8k+21k=29k$. $(5m-n)=20k-7k=13k$. Ratio = 29:13. [medium, authored]

7. If $5a + 2b = 16$ and $3a - 4b = -6$, find the value of $(a+b)^2$.

- A. 16
- B. 25 ✓**
- C. 36
- D. 9

Explanation: Multiply the first equation by 2: $10a+4b=32$. Add to the second $(3a-4b=-6)$: $13a=26 \rightarrow a=2$. Substitute back: $5(2)+2b=16 \rightarrow b=3$. $(a+b)^2=(2+3)^2=25$. [hard, authored]

8. Simplify $4\frac{2}{3} + 2\frac{1}{4} \div \frac{3}{4}$.

- A. $7\frac{2}{3}$ ✓**
- B. $6\frac{1}{3}$
- C. 7
- D. $6\frac{2}{3}$

Explanation: $2\frac{1}{4} \div \frac{3}{4} = (9/4) \div (3/4) = 3$. Then $4\frac{2}{3} + 3 = 7\frac{2}{3}$. [medium, authored]

9. A trader borrows ₹16,000 at 12% p.a. simple interest and another ₹12,000 at 18% p.a. simple interest. In how many years will the two amounts owed (principal + interest) become equal?

- A. $12\frac{2}{3}$ years
- B. $14\frac{2}{3}$ years
- C. $16\frac{2}{3}$ years ✓**
- D. $18\frac{2}{3}$ years

Explanation: This mirrors a real 2025 TNPSC Aptitude question style (same numbers, Simple Interest unit) — included here as a reminder that 'Simplification' skills (careful fraction/ratio handling) carry directly into other Aptitude sub-topics like Simple Interest. Set $16000+16000(0.12)t = 12000+12000(0.18)t$ and solve for t to get $16\frac{2}{3}$ years. [hard, question-bank-2025]

10. Find the value of: $89 \times 89 + 11 \times 11 + 2 \times 89 \times 11$.

- A. 10,000 ✓**
- B. 9,801
- C. 8,900
- D. 10,201

Explanation: This is the $(a+b)^2$ pattern in disguise: $a^2+2ab+b^2 = (a+b)^2$. Here $a=89$, $b=11$, so the answer is $(89+11)^2 = 100^2 = 10,000$ — no separate multiplication needed. [medium, authored]

Part C – General English: Vocabulary: Collocations, Compound Words, Word Creation, and British/American English

READING MATERIAL

Covers four related vocabulary skills: word pairs that naturally 'go together' (collocations), British vs. American English equivalents, how compound words are classified, and how new words form from word-parts.

Q: What is a 'collocation', and what are the natural pairings for 'rain' and 'tea' (vs. common incorrect pairings)?

A: A collocation is a pair/group of words that habitually go together in natural English, even if other word choices would seem logically similar. 'HEAVY rain' (not 'strong rain') and 'STRONG tea' (not 'heavy tea') — collocations aren't about logic, they're about convention.

Q: What are some common Verb+Noun collocations using 'make', 'do', 'take', and 'have'?

A: MAKE a decision/a mistake. DO homework/a favour. TAKE a break/a decision (note: 'take a decision' is more British usage; 'make a decision' is more universal). HAVE a break/a cold ('catch a cold' is also common). Mixing these up (e.g., 'do a decision' instead of 'make a decision') is a common error.

Q: Real exam: Which British English words were INCORRECTLY paired with their American English equivalents in a real exam question, and what are the correct pairings?

A: 'Courgette' and 'Aubergine' were swapped with their American equivalents. CORRECT pairings: Courgette (BrE) = Zucchini (AmE); Aubergine (BrE) = Eggplant (AmE). (Rocket=Arugula and Fringe=Bangs were correctly paired in that same question.)

Q: What are the four types of compound words tested in a real exam, with examples?

A: OPEN compound words: written as 2 separate words (e.g., 'ice cream'). CLOSED compound words: written as 1 single word (e.g., 'snowman'). CO-ORDINATIVE/reduplicative compounds: often rhyming or repeated-sound pairs (e.g., 'walkie-talkie'). ADJECTIVE compounds: hyphenated descriptive compounds (e.g., 'blue-eyed').

Q: Real exam: What word is formed by combining 'Vita' + 'amin'?

A: VITAMIN — confirmed by a real exam question testing word-formation/etymology (Vita = life, related to the Latin 'vita'; 'amin' relates to 'amine', a nitrogen-containing compound — together describing essential life-sustaining nitrogen compounds).

MOCK TEST (WITH ANSWERS)

1. Which is the correct collocation for describing rain that falls intensely?

- A. Strong rain
- B. Heavy rain ✓**
- C. Big rain
- D. Hard rain only

Explanation: Heavy rain. [easy, web-research]

2. Which is the correct collocation for tea with a bold, intense taste?

- A. Heavy tea
- B. Strong tea ✓**
- C. Thick tea
- D. Hard tea

Explanation: Strong tea. [easy, web-research]

3. Which verb correctly collocates with 'a decision'?

- A. Do a decision
- B. Make a decision ✓**
- C. Have a decision
- D. Get a decision done

Explanation: Make a decision. [medium, web-research]

4. What is the American English equivalent of the British English word 'Courgette'?

- A. Eggplant
- B. Arugula
- C. Zucchini ✓**

D. Bangs

Explanation: Zucchini. [hard, question-bank-2025]

5. What is the American English equivalent of the British English word 'Aubergine'?

A. Eggplant ✓

B. Zucchini

C. Arugula

D. Bangs

Explanation: Eggplant. [hard, question-bank-2025]

6. 'Ice cream' is an example of which type of compound word?

A. Closed compound

B. Open compound ✓

C. Co-ordinative compound

D. Adjective compound

Explanation: Open compound (written as 2 words). [medium, question-bank-2025]

7. 'Snowman' is an example of which type of compound word?

A. Closed compound ✓

B. Open compound

C. Co-ordinative compound

D. Adjective compound

Explanation: Closed compound (written as 1 word). [medium, question-bank-2025]

8. Combining 'Vita' and 'amin' forms which word?

A. Vaccine

B. Vitamin ✓

C. Vitality

D. Vitalize

Explanation: Vitamin. [medium, question-bank-2022]